

Data Analysis Project: Coffee Vending Machine

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*Picture just illustration



Agenda

- Problem Statement
- About Data
- Key Findings
- Suggestions



Problem Statement

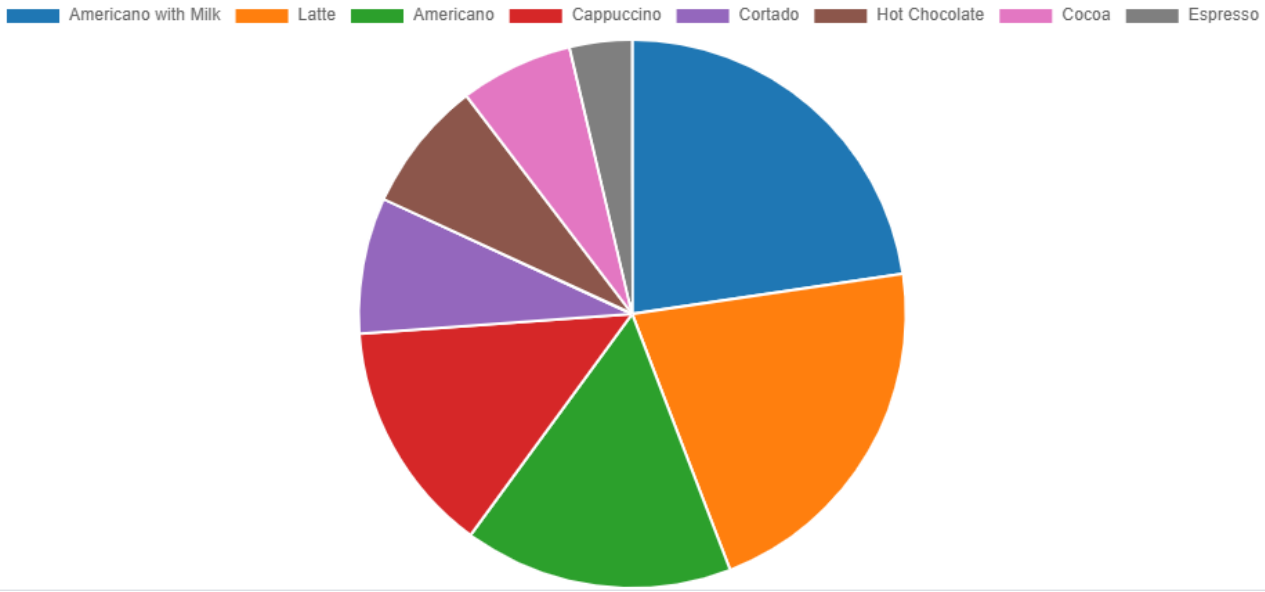
- Dummy dataset are from Kaggle (by Yaroslav Isaienkov @ihelon) contains records of coffee sales from vending machine and added self-made dummy data.
- To narrow the context, I assume that the vending machine was put at the lobby of an university.
- By analyzing the data, my goal is to find insight that may help with sales boosting.



About Data

- Coffee Vending Machine data
- Data Coverage: March 2024-March 2025
- Data Size: 561 KB, 6 columns, 3636 rows
- Added Data Size: 45 KB, 3 columns, 1317 rows
- Fields: Date, datetime, transaction type, card code (for card type transactions), price, coffee name, first and last name (on added dataset only)

Key findings I: Coffee sales by numbers, revenue, and date



	Coffee Name text	total_orders bigint
1	Americano with Milk	824
2	Latte	782
3	Americano	578
4	Cappuccino	501
5	Cortado	292
6	Hot Chocolate	282
7	Cocoa	243
8	Espresso	134

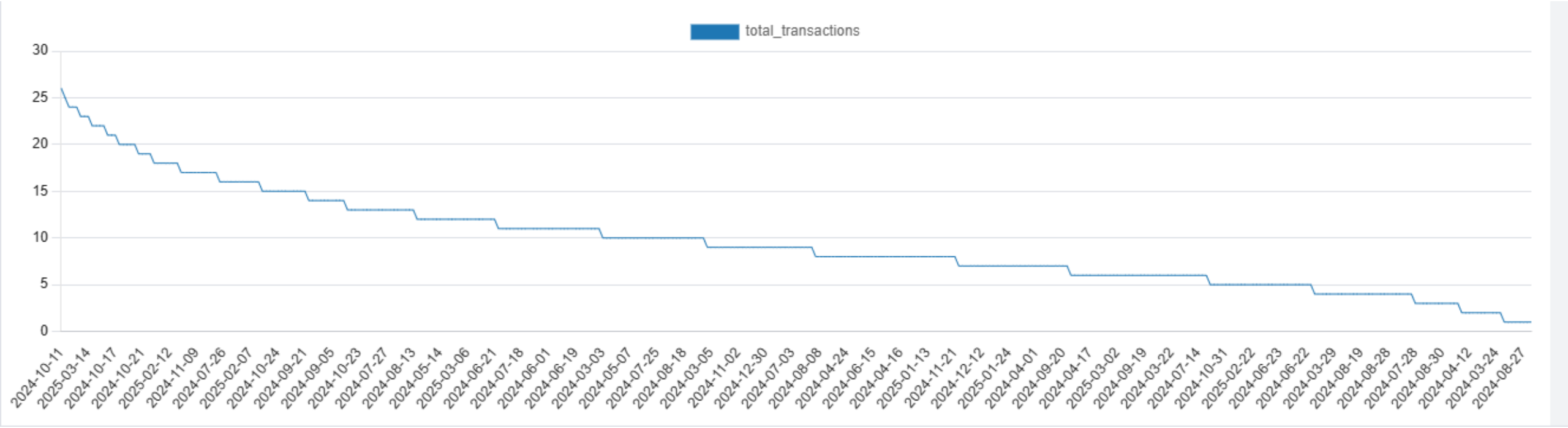
	Coffee Name text	revenue numeric
1	Latte	27866.30
2	Americano with Milk	25269.12
3	Cappuccino	18034.14
4	Americano	15062.26
5	Hot Chocolate	10172.46
6	Cocoa	8678.16
7	Cortado	7534.86
8	Espresso	2814.28

Americano with Milk sold the most but **Latte has the most revenue.** Americano with Milk is cheaper than Latte by 4,9.

Difference between both product are the taste. Americano with milk taste lighter and more coffee-ish and Latte taste creamier and milkier.

By price, Americano is cheaper. This may encourage more purchase among the customers.

Key findings I: Coffee sales by numbers, revenue, and date

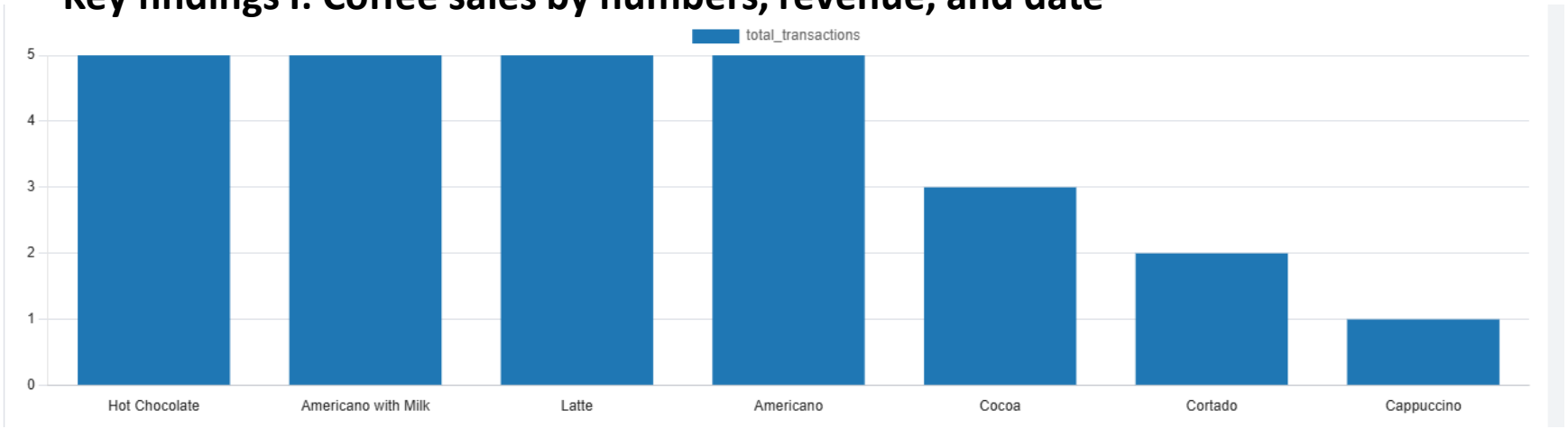


	Date date	total_transactions bigint
1	2024-10-11	26
2	2025-02-03	25
3	2024-09-22	24

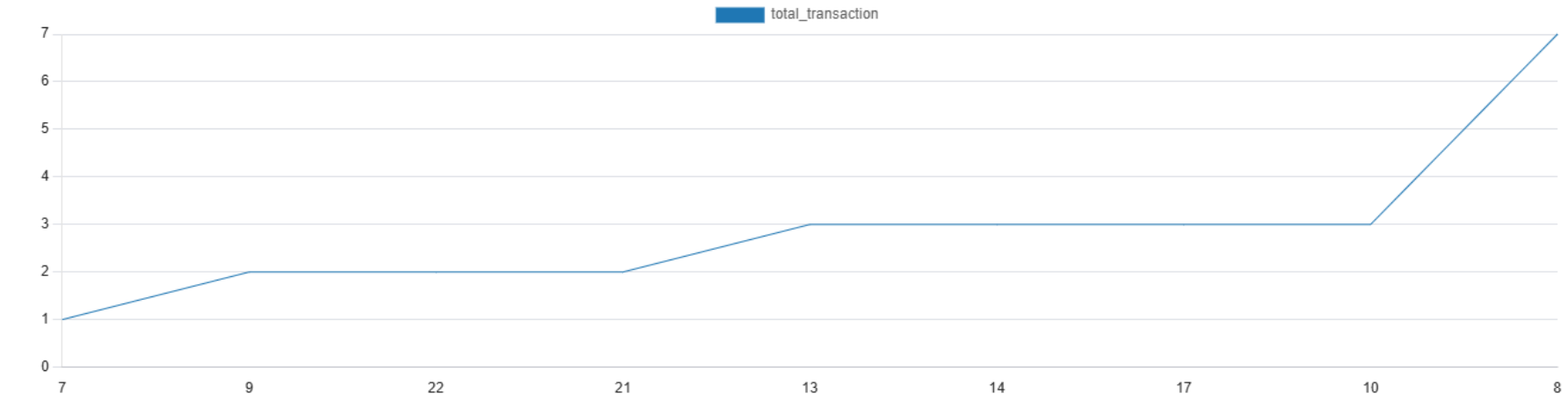
	Date date	total_revenue numeric
1	2024-10-11	836.66
2	2025-02-03	796.00
3	2025-02-05	760.24

The graphic shows the date with most purchase to least purchase, **not gradually lower sales.** From top three date by total transactions and total revenue 11 October 2024 and 3 February 2025 is the consistent data. Since the gap between 11 October 2024 and 3 February 2025 is quite big, we are going to **focus on 11 October 2024 Sales.**

Key findings I: Coffee sales by numbers, revenue, and date



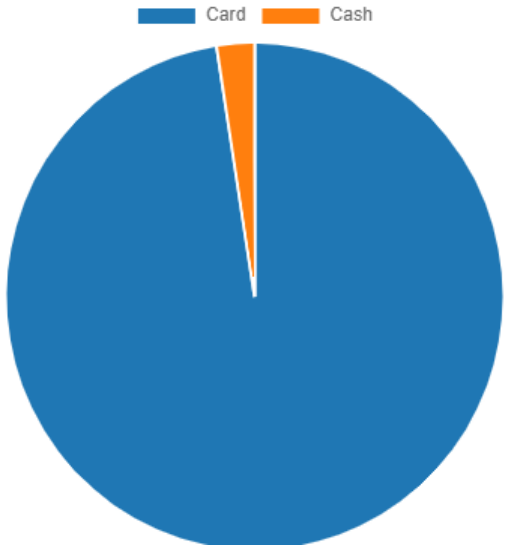
Sales by coffee name



Sales by hour. X line shows hour, Y line shows sales number

11 Oct '24 has the most spending with total spent **26 transaction** and **836.66 as total revenue**. Hot Chocolate, Americano, Americano with Milk and Latte sold the most that day. **Mostly people buy drinks from the coffee vending machine at 8 A.M.**

Key findings II: Customers’ behavior by transaction type



	Card Code text	total_purchases bigint
1	ANON-0000-0000-0012	129
2	ANON-0000-0000-0141	108
3	[null]	89

By the chart, **97.55%** of customers doing their transaction with card.
Top three customers: ANON-0000-0000-0012, ANON-0000-0000-0141, and [null]. The data [null] means cash transactions, it doesn't refer to a person because it could be purchased by different people.

By cards transactions, it’s easier for the customer to pay and it’s easier for us to trace them by their names and how much they have purchased.

Key findings II: Customers' behavior by transaction type

	Card Code text	First Name text	Last Name text	total_spending numeric
1	ANON-0000-0000-0012	Liam	Gonzalez	3785.92
2	ANON-0000-0000-0141	Liam	Scott	2749.78
3	ANON-0000-0000-0009	Emma	Roberts	2343.98

Liam Gonzalez, Liam Scott, and Emma Roberts are top three customers can be trace by their card code.

Dataset added with second dataset for tracing the customers who did card transactions type. By knowing this data, we can give discount for top spender customer. In this case Gonzalez, Scott, and Roberts may have their discount.



Suggestions

- By key findings I, stock more Americano with Milk and Latte will be more beneficial.
- 11 October 2024 has the most purchased number. This could have many factors; early autumn season (that is why Hot Chocolate appeared at the top sale), start of mid-term exam, or maybe there are an event there. This one needs more data to make sure.
- 8 A.M also has the most purchased number. It means customers will buy coffee before they do their activity.
- By key findings II, 97.55% of the customer doing card type transactions, we can trace them by their card and we can give them discounts for the top spender.